25901192 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECSC200 - DATA MINING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define decision tree. [U] / [CO1]
2. Define lift in correlation. [U] / [CO2]
3. Differentiate between Single, Complete, and Average Linkage Methods. [An] / [CO3]
4. Describe methods to handle missing values. [U] / [CO1]
5. State the role of clustering in real-world applications. [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Discuss about association analysis. [U] / [CO1]
7. Differentiate between OLTP and OLAP. [U] / [CO2]

8. Compare SVM and KNN classification methods. [U] / [CO3]

9. Illustrate how Data Mining can be applied in the banking or retail sector. [A] / [CO1]

10. Explain the major issues in Data Mining with suitable examples. [U] / [CO1]

11. Explain four widely used measures for distance between Clusters in hierarchical clustering. [U] / [CO3]

12. Explain fact table and dimension table. [U] / [CO2]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Explain Association Rule Mining with support and confidence. [U] / [CO2]

14. Describe the Naïve Bayes Classification algorithm with suitable example. [U] / [CO3]

[15x1 = 15]